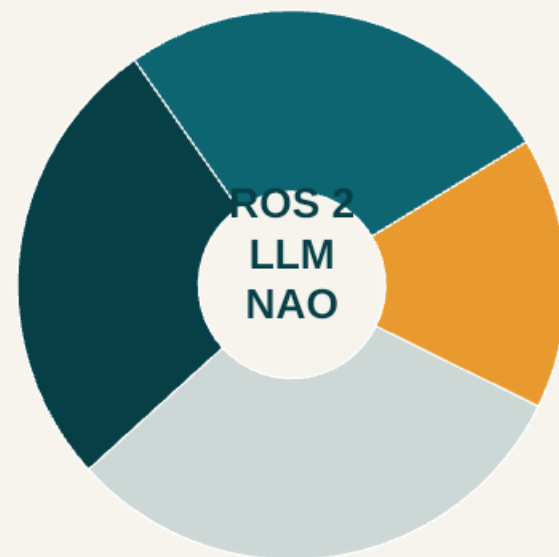


Design, Integration, and Validation of a Modular LLM-Based Interactive Architecture

A NAO-centred ROS 2 architecture for grounded dialogue, structured planning, deterministic execution, and evidence-based reporting.

Juan David Bendek Williamson • Supervised by Raquel Ros • 2025–2026





A plausible plan is not yet a safe robot action.

Language is flexible

- Users refer to people, objects, and prior turns.
- One request can mix dialogue, knowledge, and action.
- A fluent plan can still omit targets or invent unsupported behaviour.

Robots require evidence

- Capabilities and arguments must be checked before dispatch.
- Identity, order, and lineage must survive execution.
- Completion must follow typed results and verified effects.

Example: “Bring all the objects you can see to ALEX.”



The study asks how flexibility can be integrated without surrendering execution control.

How can a modular robot architecture integrate dialogue, symbolic grounding, planning, deterministic execution, and feedback while preserving clear responsibility boundaries and evidence-based communication?

Integrate

dialogue, grounding, planning, orchestration, and skills through explicit ROS 2 interfaces

Constrain

candidate plans through reviewed capabilities and deterministic admission before dispatch

Validate

full turns, controlled failures, symbolic postconditions, and truthful reporting

RQ1 plans

RQ2 grounding

RQ3 admission

RQ4 recovery

RQ5 ownership



The architecture contains model variation instead of assuming it away.

1

Structure

Generated outputs enter typed request and plan contracts.

2

Validate

Unsupported skills, roles, arguments, and incomplete coverage are rejected.

3

Execute

The orchestrator controls ordered dispatch and plan lineage.

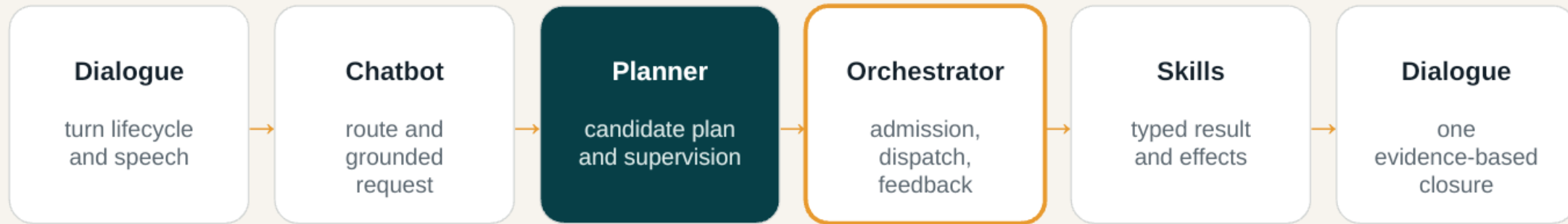
4

Report

Typed results and postconditions constrain user-facing claims.

Deterministic interfaces do not make generation deterministic. They make variation attributable and restrict its execution authority.

Each uncertainty has an owner, and evidence returns through the same lineage.



Goal ID • plan ID • plan version • stable step IDs

Symbolic evidence

KnowledgeCore and scene grounding provide bounded entities, roles, locations, and relations.

Execution evidence

Robot-facing or simulated adapters return the same typed result boundary.



Explicit interfaces separate proposal quality from execution safety.

Planner proposes

- structured goal and target selection
- ordered candidate steps
- retry or replan policy

Admission decides

- registered skill and arguments
- role and target-set integrity
- coverage, dependencies, and lineage

Executor dispatches

- only accepted steps
- typed status and effects
- correlated feedback

Safe rejection prevents invalid dispatch, but it does not convert an incomplete plan into success.



Available context supports a decision, but fresh evidence proves an action.

Role-preserving symbolic context

person → recipient or human target
object → manipulable entity
location → navigation anchor
support → object relation

requested → selected → planned → executed →
reported

Robot-facing in v1

Speech, posture and head motion were integrated.
Look-at used a real adapter, while the primary profile remained fake-by-default.

Controlled adapters in v1

Navigation, find, pick, place, bring, walk and wave exposed deterministic success and failure outcomes through the same result contract.

The detector and physical manipulation remain separate evidence seams.



A case passed only when semantics, execution, effects, and reporting agreed.

A

Semantic route

Correct dialogue,
knowledge, or execution
path

B

Accepted plan

Complete targets, roles,
skills, order, and lineage

C

Controlled outcome

Success, failure,
clarification, or bounded
replan

D

Verified closure

Postcondition and final
report agree with
evidence

One unit of analysis = one user request under a fixed software, model, world-state, and skill configuration.

Scenario-based software and integration validation, not unrestricted robot-task success or human-subject evaluation.



One primary qualification anchors the thesis; other model probes remain separate.

108

Retained case records

Derived from 33 structured runtime traces across the final evidence index.

72

Primary records

Qwen3-VL: 64 passes and 8 failures under the frozen qualification.

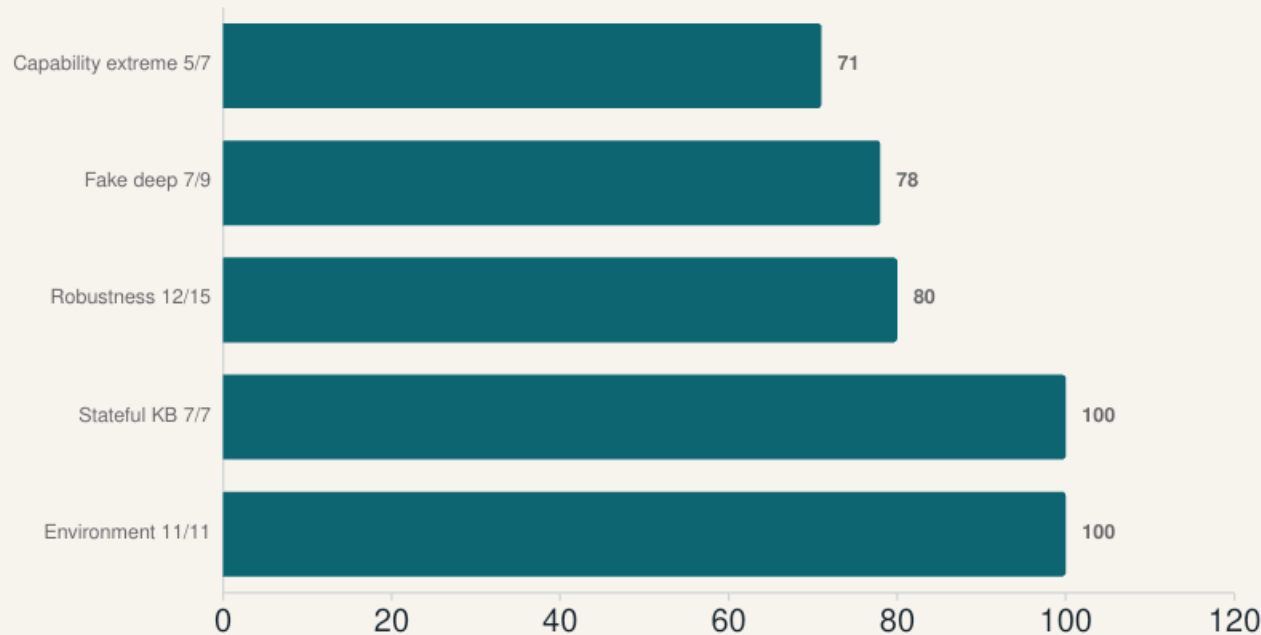
191

Model supplement

Three Ollama models, retained outside the primary denominator.

The 22-case end-to-end questionnaire produced 20 passes and 2 failures.

Most tested seams passed. Semantic non-passes remained visible.



What the failures exposed

- incomplete universal target selection
- one location-to-recipient role error
- missing terminal closure after recovery
- incomplete multi-capability plan coverage

Interpretation

Incomplete plans were rejected before dispatch. Safe containment did not erase task-level failure.



The remaining failures occur at identifiable semantic and closure boundaries.

ROUTE	KB update entered the execution planner	Route and ownership failure
SELECTION	One member omitted from an “all objects” set	Target-set failure
ROLE	Kitchen treated as the delivery recipient	Entity-role failure
COVERAGE	Required action absent from a long composite	Plan-completeness failure
CLOSURE	Recovery occurred without reliable terminal speech	Report/closure failure

The architecture converted silent execution risks into inspectable route, selection, plan, clarification, or closure failures.



Stable interfaces did not produce invariant semantic behaviour.

Stable across backends

- Environment profile
- Startup ownership
- Symbolic transport
- Planner request schema
- Typed execution boundary

Variable across backends

- Route selection
- Target coverage
- Role assignment
- Clarification
- Plan and report completeness

Qualification rule

Qwen3-VL completed the frozen campaign. A compatible replacement must pass the same campaign before inheriting qualification.

Model-agnostic describes the interface boundary. Model-invariant would require equivalent behaviour, which the evidence did not show.

A failed pick produced a revised plan without losing the completed work.



The model did not directly retry the robot. It proposed plan version 2 after receiving structured failure evidence.

Why this trace matters

It demonstrates middle-stage recovery, retained lineage, and a truthful closure. Beginning-stage navigation recovery was also retained; terminal recovery remained less reliable.



Three short turns expose dialogue, knowledge, and execution routing.

01

“Hey, how are you?”

one response, no planner request

Dialogue route

02

“What can you see now?”

answer from the declared symbolic state

Knowledge route

03

“Move your head up.”

accepted plan, one skill result, one closure

Execution route

The demonstration illustrates the architecture. The frozen traces remain the scored evidence.



The strongest evidence concerns software contracts and symbolic state.

Supported

- ownership and interface architecture
- deterministic admission and dispatch
- controlled failure and replanning
- symbolic postconditions and reporting

Not established

- navigation or grasp accuracy
- detector performance
- human safety or acceptance
- unrestricted autonomy or universal model ranking

Transfer conditions

Another robot needs an equivalent reviewed skill registry and typed adapters. Another model must repeat the frozen qualification.

Controlled simulation reduced hardware variance and compute waste, but it cannot replace physical safety and performance measurement.



The thesis result is preserved as one reproducible configuration.

v1

Code

Reviewed branch and
nested fork commits.

v34

Container

iiia:nao-final

Q3VL

Primary model

Qwen3-VL 30B A3B
AWQ

8.4

Operational index

Secondary summary of
the explicit case
evidence.

sha256:bb82c158092a69870dae48272b3b20fd8cd4ecfe97a0e61a2253a2cca00c663e

The score summarises qualification; the denominators and failure reasons carry the claim.



Move from validated software contracts toward measured embodiment.

Embodied skills

Quantify head and gaze behaviour, guarded wave, bounded local motion, semantic navigation, and pointing kinematics.

World evidence

Evaluate detector identity settling and scene-grounding quality before promoting detector-derived facts into stronger claims.

Interaction quality

Measure ASR, TTS, latency, interruption handling, and autonomous idle/request arbitration with users and physical trials.

Every promoted skill should begin with one physical success, one failure or cancellation, and the same typed evidence boundary.



The thesis contributes an architecture, a containment boundary, and a qualification method.

01

Architecture

A modular ownership model connecting dialogue, grounding, planning, execution, and reporting.

02

Containment

Typed contracts and deterministic admission restrict model-generated execution authority.

03

Qualification

A repeatable evidence method separates model, stack, observability, and physical claims.

Together, these boundaries make language-model variation observable without hiding task-level failure.



Language-model dialogue and planning can participate in an integrated robot architecture when their outputs are grounded, validated against reviewed skills, dispatched deterministically, and checked against typed feedback and symbolic postconditions.

The boundaries do not eliminate non-determinism. They make it observable and restrict its execution authority.

Thank you. I welcome your questions.



Detailed frozen results

Evidence scope	Pass	Fail	Interpretation
Primary qualification	64	8	72 frozen Qwen3-VL records
End-to-end questionnaire	20	2	22 integrated user turns
Environment	11	0	Controlled scene and grounding
Stateful KB	7	0	Pre-query, mutation/effect, post-query
Robustness	12	3	Five frozen cases repeated three times
Fake deep	7	2	Nine demanding multi-step cases
Capability extreme	5	2	Long composite coverage



Research-question synthesis

RQ1	Supported for tested task families	One questionnaire composite and two extreme cases failed
RQ2	Strong in controlled symbolic profiles	Selection and role assignment remained model-sensitive
RQ3	Supported by deterministic rejection	Tested schemas, arguments, lineage, and registry only
RQ4	Direct support for fail-once navigation and pick	Missing-object, recipient, and closure recovery remained partial
RQ5	Supported at interface and auditability level	Other robots and equivalent model behaviour remain unqualified

Frozen runtime tuple



Image	iiia:nao-final	sha256:bb82c158092a...
Primary model	Qwen3-VL 30B A3B AWQ	chatbot, intent, planner
Pipeline	response-first	JSON-only grounded context
Primary evidence	72 records	64 pass, 8 fail
Detector	disabled in primary profile	separate perception scope
Physical scope	speech, posture, head/gaze integration	no quantified navigation or manipulation

Demonstration operating rule

Proceed live only if preflight is green

- one model endpoint advertised and probed
- clean stack with unique nodes
- rqt input and trace viewer visible
- declared KB fixture verified
- robot adapter selected before the defence

Otherwise use retained evidence

- name the unavailable dependency once
- do not relaunch or switch models on stage
- show the representative trace slide
- state that the live run is illustrative and the retained campaign is scored

Maximum live window: 90 seconds. No troubleshooting inside the timed defence.



Short answers for likely questions

Why simulated skills?

They place repeatable outcomes at the robot-facing result boundary, isolating planning and recovery from hardware variance.

What is deterministic?

Admission, schema checks, dispatch order, lineage, feedback correlation, and effect verification. Route and plan generation remain model-dependent.

Why model-agnostic?

The interfaces accept multiple backends. The results explicitly show that semantic behaviour is not invariant.

Why no classical planner baseline?

The thesis validates integration and containment, not a causal comparison of planning algorithms.

What comes next?

Quantified physical skills, detector-quality evaluation, and user-facing interaction trials under the same evidence contracts.



Full implementation and validation objectives

1

Integrate dialogue, routing, grounding, planning, deterministic orchestration, and skills through ROS 2 interfaces.

2

Define explicit runtime contracts for requests, context, plans, results, feedback, dialogue acts, and raw evidence.

3

Expose capabilities through a reviewed skill registry rather than model-generated robot APIs.

4

Support dialogue, knowledge, atomic, and composite turns without transferring speech or execution ownership.

5

Validate plans before dispatch and return typed evidence for failure, clarification, replanning, and reporting.

6

Exercise full turns, controlled outcomes, focused subsystem checks, and symbolic postconditions.